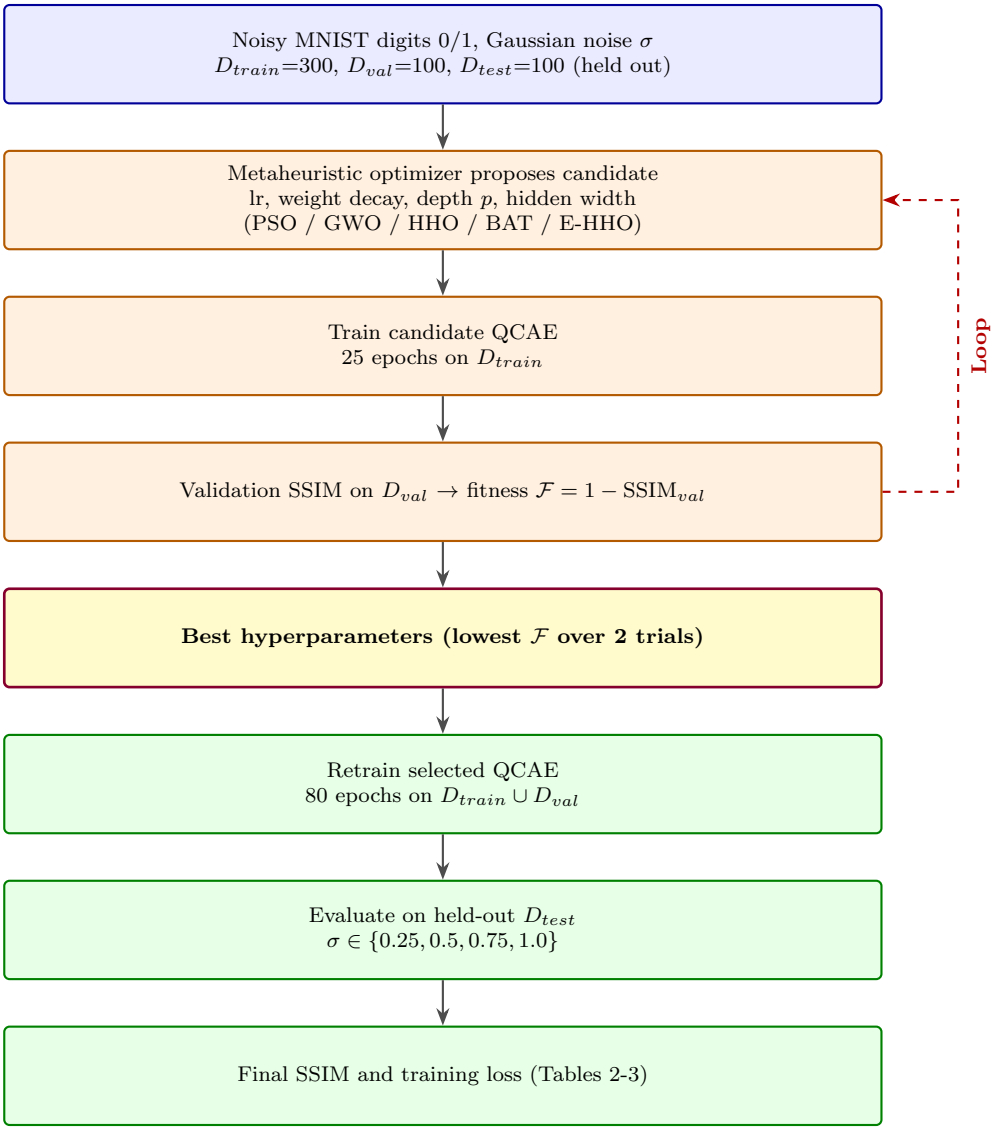

Noisy MNIST digits 0/1, Gaussian noise σ
D_{train}=300, D_{val}=100, D_{test}=100 (held out)
Metaheuristic optimizer proposes candidate
lr, weight decay, depth p, hidden width
(PSO / GWO / HHO / BAT / E-HHO)
Train candidate QCAE
25 epochs on D_{train}
Validation SSIM on D_{val} → fitness $\mathcal{F} = 1 - \text{SSIM}_{val}$
Loop
Best hyperparameters (lowest $\mathcal{F}$ over 2 trials)
Retrain selected QCAE
80 epochs on $D_{train} \cup D_{val}$
Evaluate on held-out D_{test}
$\sigma \in \{0.25, 0.5, 0.75, 1.0\}$
Final SSIM and training loss (Tables 2-3)